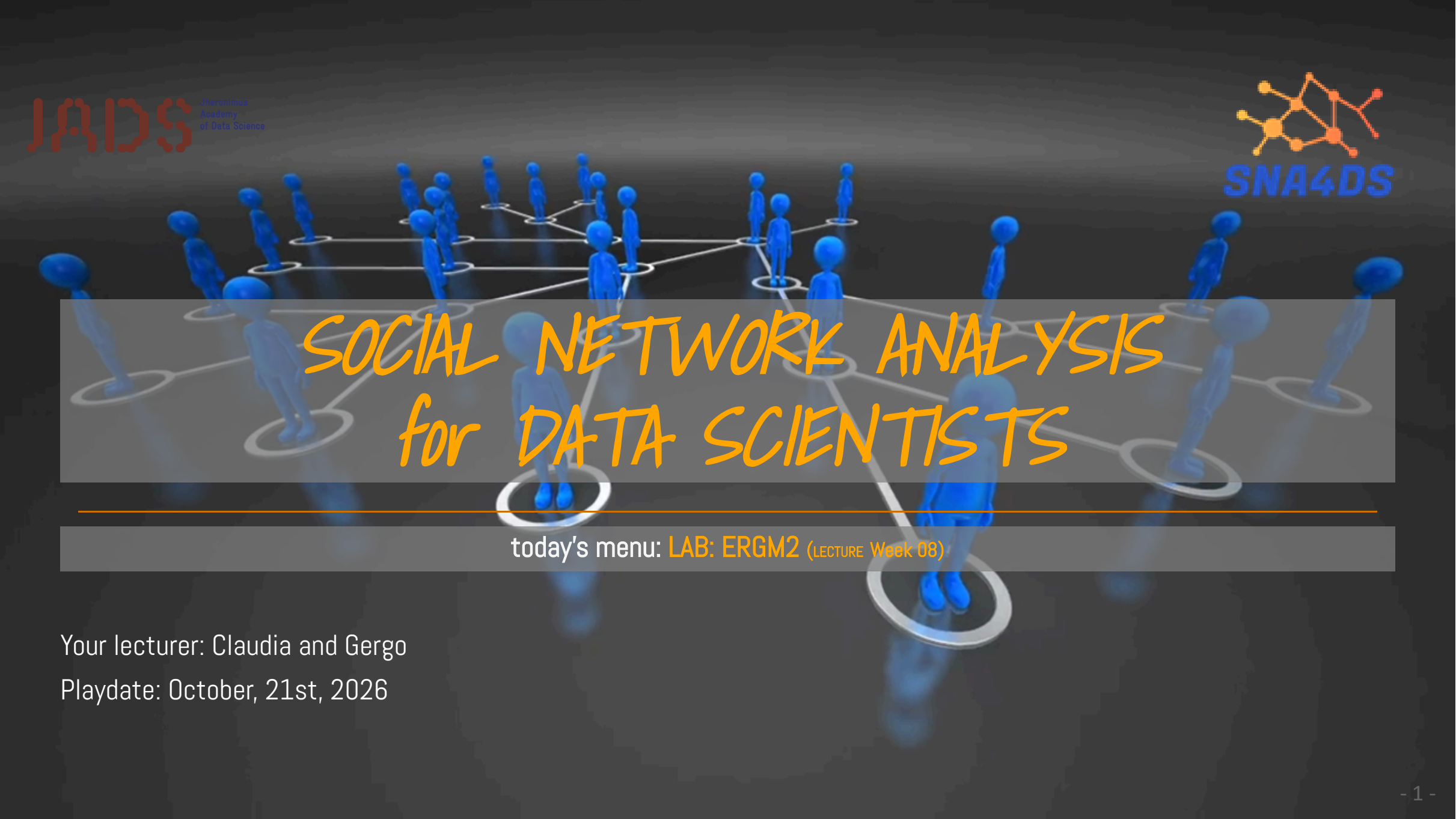


SOCIAL NETWORK ANALYSIS for DATA SCIENTISTS

today's menu: LAB: ERGM2 (LECTURE Week 08)

Your lecturer: Claudia and Gergo

Playdate: October, 21st, 2026

Menu'

- ERGM terms bootcamp
- The Intensity of the Effects
- 7 steps Research Flow

ERGM terms bootcamp

Wooclap

SNA4DS

The term test!

MKRQLBY

Counting ERGM terms

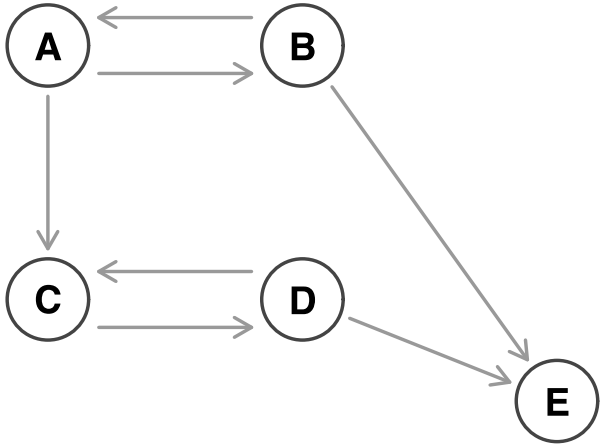
Count each statistic, then check the highlighted answer.

All ties are directed. An **edge** is one arrow; a **dyad** is one unordered node pair.

Edges: count

How many arrows? Count both directions of a reciprocal pair.

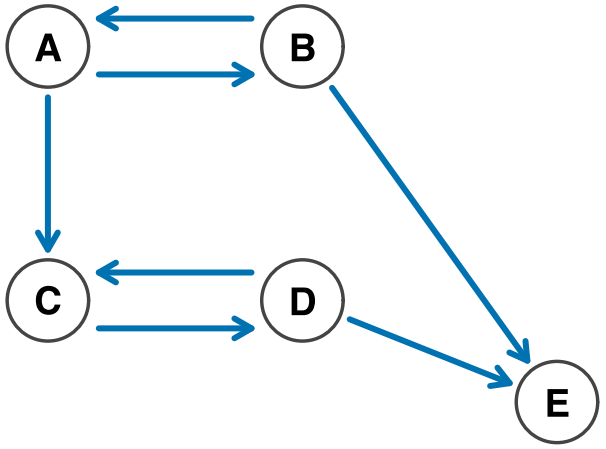
`edges` = ____



Edges: answer

Each arrow counts once.

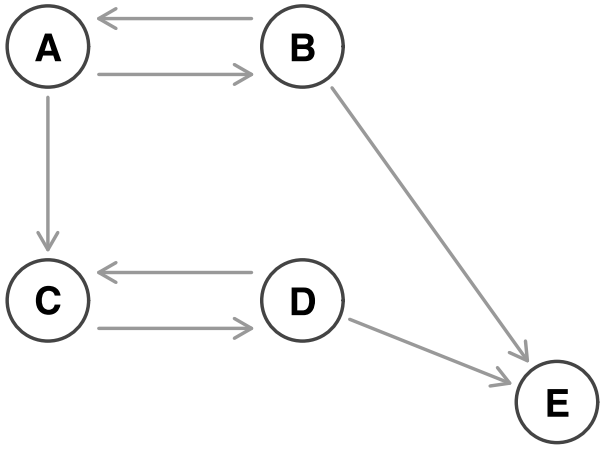
`edges` = 7



Mutual dyads: count

How many pairs have arrows in both directions? Name them.

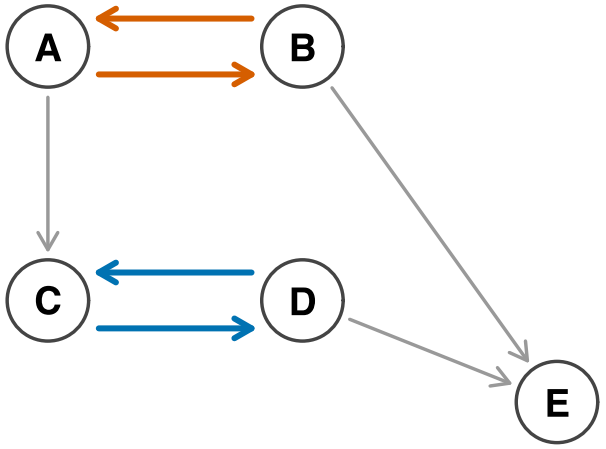
`mutual` = ____



Mutual dyads: answer

A–B and C–D. Each pair counts once.

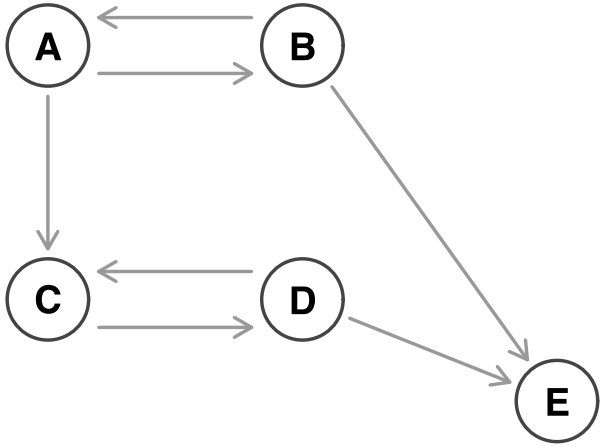
`mutual` = **2**



Two-out-stars: count

A **2-out-star** is two arrows with the same sender. Count the stars and identify their arrows.

`ostar(2)` = ____

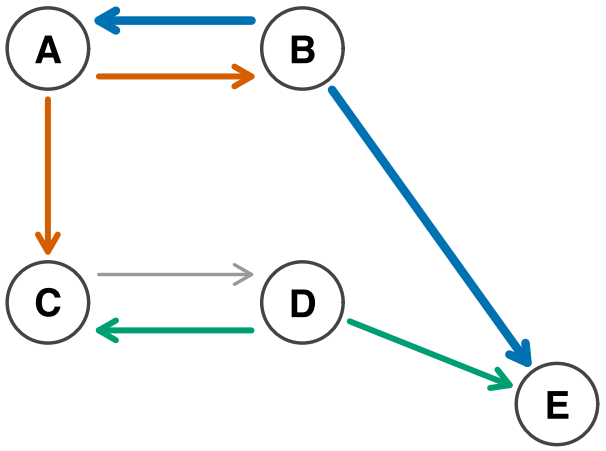


Outdegree d contributes $\binom{d}{2}$ stars.

Two-out-stars: answer

A, B and D each contribute one star.

`ostar(2)` = 3

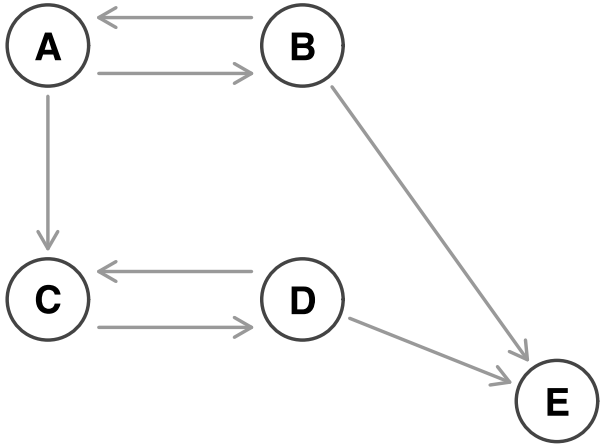


A→B / A→C; B→A / B→E; D→C / D→E. Each color marks one star.

Two-in-stars: count

A **2-in-star** is two arrows with the same receiver. Count the stars and identify their arrows.

`istar(2)` = ____

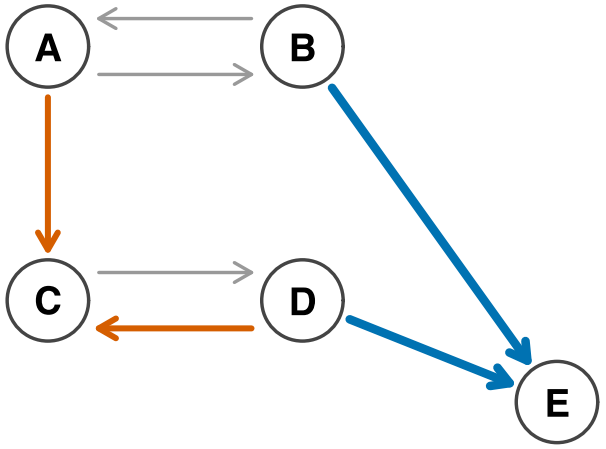


Indegree d contributes $\binom{d}{2}$ stars.

Two-in-stars: answer

C receives from A and D; E receives from B and D.

`istar(2)` = 2

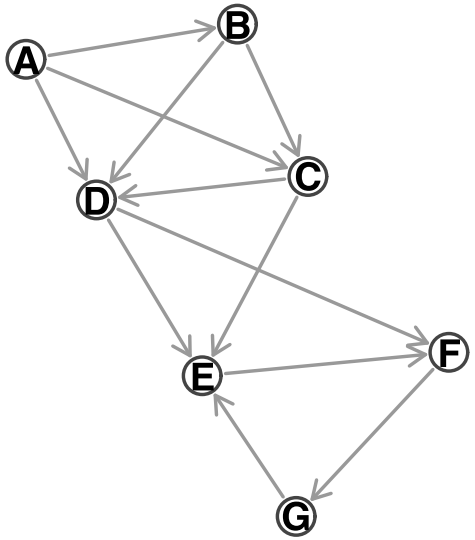


Each color marks one star; grey arrows belong to none.

Closed triads: count

No reciprocal ties. List triples with all three pairs connected (`triangle`); identify directed cycles (`ctriple`).

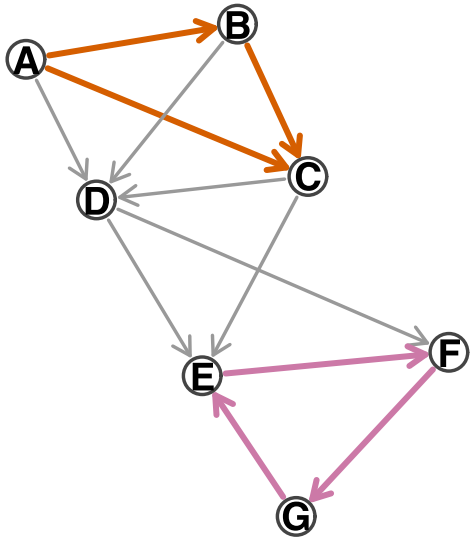
`triangle` = ____ · `ctriple` = ____



Closed triads: answer

Orange: transitive ABC. Purple: cyclic EFG.

`triangle` = 7 · `ctriple` = 1

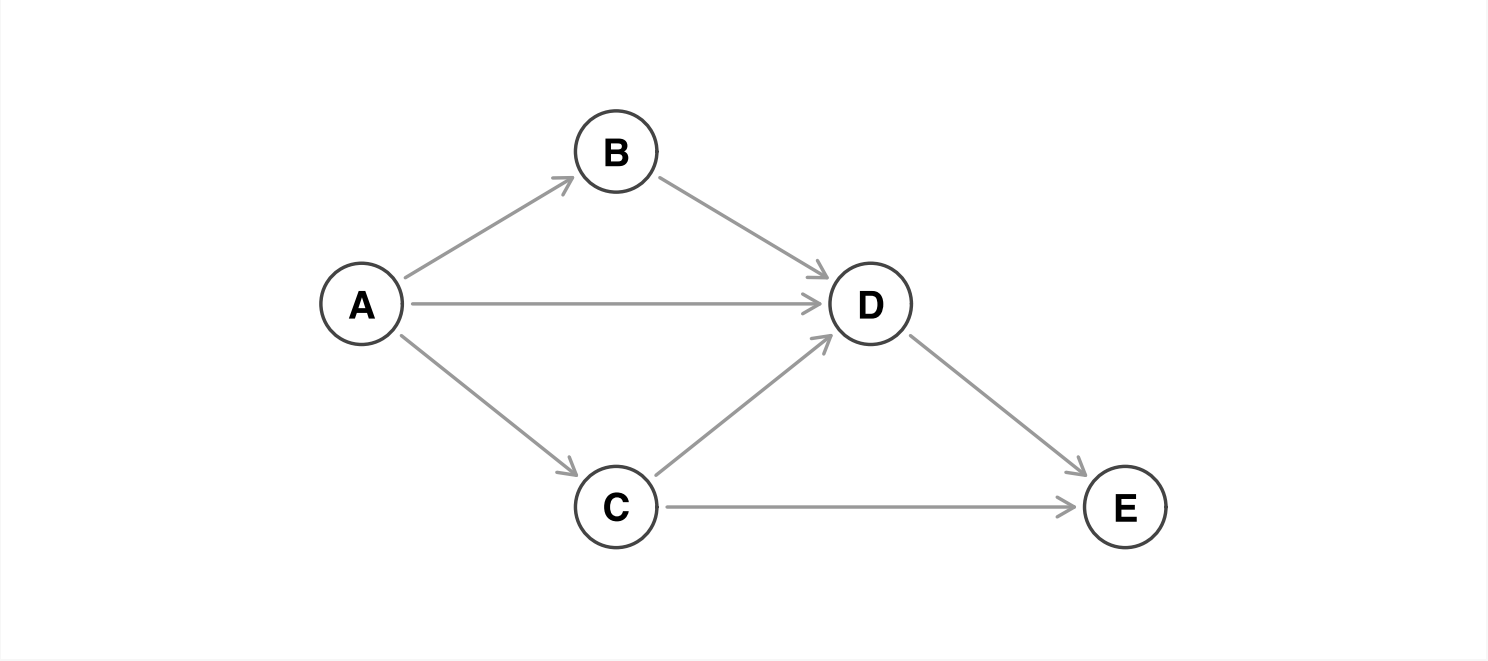


Transitive: ABC, ABD, ACD, BCD, CDE, DEF. Cyclic: EFG.

Edgewise shared partners: count

For **OTP**, a shared partner of $i \to j$ is a node k on $i \to k \to j$. Which arrows have one partner? Two?

`esp(1)` = ____ · `esp(2)` = ____

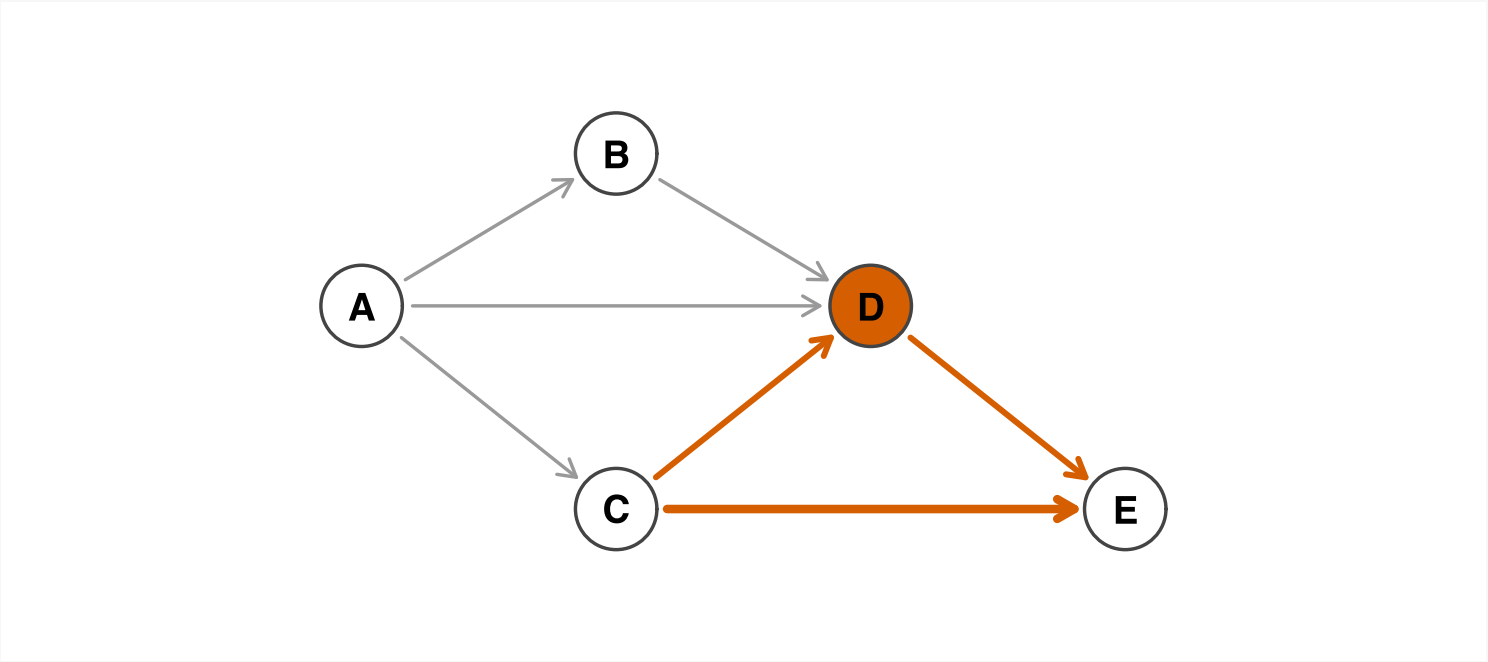


`esp(d, type="OTP")` counts arrows with exactly `(d)` outgoing two-path shared partners.

One shared partner: answer

C→E has one partner: D, via C→D→E.

`esp(1)` = 1 · `esp(2)` = ____

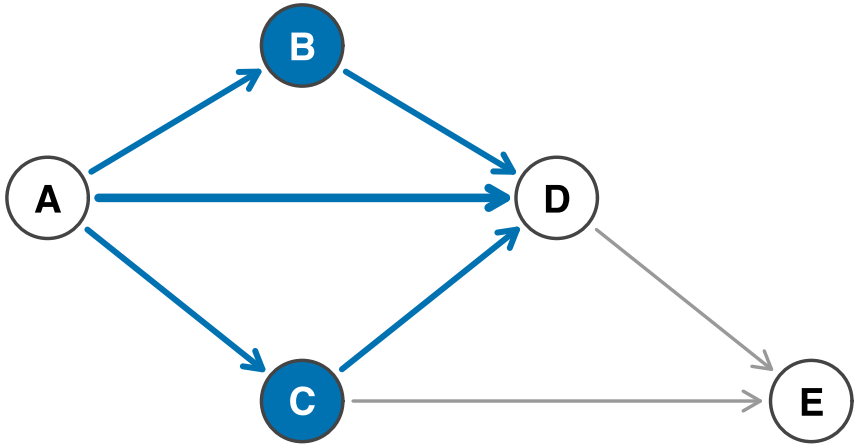


The thick arrow is counted; the other colored arrows show the two-path.

Two shared partners: answer

A→D has two partners: B and C.

`esp(1)` = 1 · `esp(2)` = 1



A→B→D and A→C→D. The target arrow contributes **one** to `esp(2)`, not two.

From ESP to GWESP

With decay 0.5, an arrow with k OTP shared partners contributes

$$w(k) = e^{0.5}\left[1 - \left(1 - e^{-0.5}\right)^{k}\right].$$

Arrow	Partners	ESP statistic	GWESP contribution
C→E	D	`esp(1) = 1`	$w(1) = 1$
A→D	B, C	`esp(2) = 1`	$w(2) = 2 - e^{-0.5} \approx 1.393$

`gwesp(0.5, fixed=TRUE, type="OTP")` = **2.393**.

The first partner adds 1; the second adds about 0.393. GWESP combines ESP counts with diminishing contributions from additional partners.

Counts and coefficients

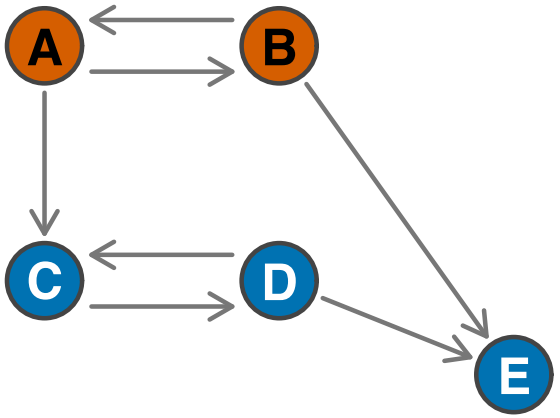
These counts are the ERGM's sufficient statistics $g(y)$:

- **Network 1:** `edges` = 7, `mutual` = 2, `ostar(2)` = 3, `istar(2)` = 2.
- **Network 2:** `triangle` = 7, `ctriple` = 1.
- **Network 3:** `esp(1)` = 1, `esp(2)` = 1, `gwesp(0.5)` = 2.393.

Coefficients weight these statistics in the model. Toggling one arrow can change several statistics at once.

Estimation tricks: checkpointing

Five nodes, seven directed ties.



Orange: group one (A/B).

Blue: group two (C/D/E).

Overwritten at the start of each MCMLE iteration.

```
checkpoint_file <- "ergm-checkpoint.RData"

fit <- ergm::ergm(
  network ~ edges + mutual + nodematch("group"),
  control = ergm::control.ergm(
    MCMC.samplesize = 500,
    MCMC.burnin = 100,
    MCMLE.maxit = 5,
    parallel = 0,
    parallel.type = "PSOCK",
    checkpoint = checkpoint_file,
    resume = if (file.exists(checkpoint_file)) checkpoint_file
  )
)
```

```
Iteration 1 of at most 5:
Saving state in 'ergm-checkpoint.RData'.
```

Estimation tricks: resuming a fit



If RStudio crashes or you stop the fit, rerun the whole code block. The fit resumes from the last checkpoint.

Output after rerunning:

```
Resuming from state saved in 'ergm-checkpoint.RData'.
Iteration 1 of at most 5:
Saving state in 'ergm-checkpoint.RData'.
```

The counter restarts at 1, but the fit uses the saved state. Use the same network and model.

To start again, delete the checkpoint:

```
file.remove("ergm-checkpoint.RData")
```

```
[1] TRUE
```

The Intensity of the Effects

Interpreting model results

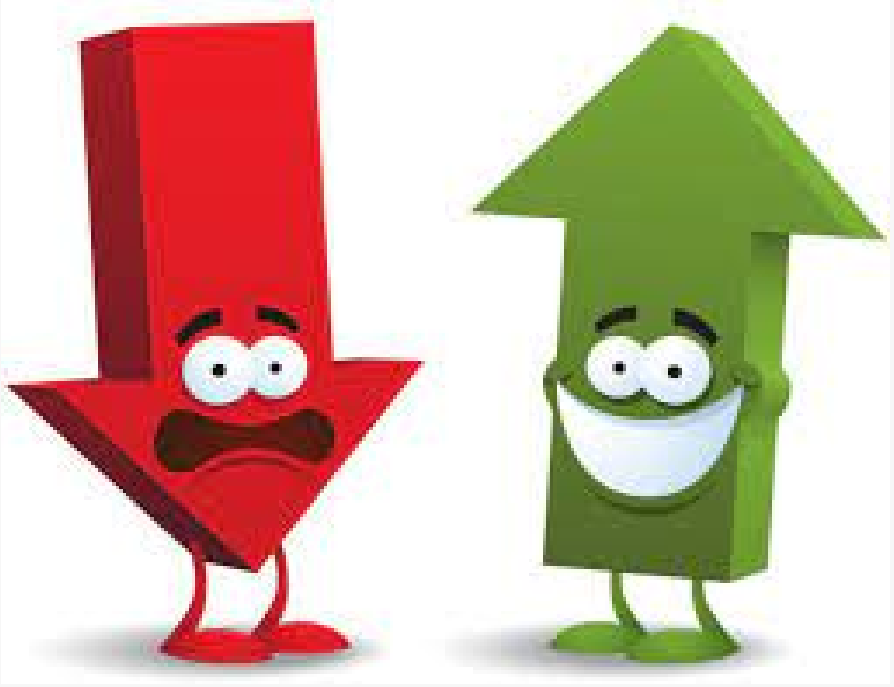
1. $\theta > 0$ and stat. signif. -> X increases P(1) and decreases P(0)
2. $\theta < 0$ and stat. signif. -> X decreases P(1) and increases P(0)
3. θ not stat. signif. -> X does not affect P(1) and P(0)

Binary

The sign of θ provides us with a binary interpretation:

the probability of the edge forming

- Increases
- Decreases



Ok, sure... How much!?

The number tells us "how much'' ---- Intensity



How much … or "The extent to which…"

x predicts y

- The odds that x predicts (y) edge formation
- The probability that x predicts (y) edge formation

Odds ratios (Memo: Bootcamp!)

ERGMs work with log scale, right?

Well, to interpret our coefficients we need to exponentiate it. Makes sense, doesn't it?

When you exponentiate you estimate the odds of something to happen --

ODDS = probability of success

FORMULA

$$OR = exp(\theta)$$

Odds Ratios meaning (Memo: Bootcamp!)

If the coefficients are statistically significant it is possible to comment on the odds of the effect taking place

Interpretation:

An increase of 1 in X_1 increases the log odds with θ_1 (log odds) and it makes the odds of the effect taking place $exp(\theta_1)$ times larger.

Consider that...

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Odds Ratio is a measure of the strength of association with an exposure and an outcome.

1. OR > 1 means greater odds of association with the exposure and outcome.
2. OR = 1 means there is no association between exposure and outcome.
3. OR < 1 means there is a lower odds of association between the exposure and outcome.

Probabilities

It measures the probability of an effect taking place

FORMULA:

$$P = \frac{exp(\theta)}{1+exp(\theta)}$$

WARNING: if you want to use % the result needs to be multiplied by 100.

Back to Florence

Let's see the intensity of some effects



Dyadic independent model

```
m <- ergm::ergm(flom ~ edges + nodecov("Wealth"))
```



Coefficients

θ

```
summary(m)
```

```
Call:
ergm::ergm(formula = flom ~ edges + nodecov("Wealth"))

Maximum Likelihood Results:

                Estimate Std. Error MCMC % z value Pr(>|z|)
edges          -2.594929   0.536056      0  -4.841   <1e-04 ***
nodecov.Wealth  0.010546   0.004674      0   2.256   0.0241 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

     Null Deviance: 166.4  on 120  degrees of freedom
 Residual Deviance: 103.1  on 118  degrees of freedom

AIC: 107.1  BIC: 112.7  (Smaller is better. MC Std. Err. = 0)
```

Coefficient interpretation

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Edges -2.595

$\theta < 0$ and stat. signif. -> X decreases P(1) and increases P(0)

Wealth 0.011

$\theta > 0$ and stat. signif. -> X increases P(1) and decreases P(0)

We still don't know anything about the intensity!!!

Odds Ratios

$exp(\theta)$

```
snafun::stat_ef_int(m, type = "odds")
```

```
               Estimate  Odds Std.Error  Pval
edges            -2.595 0.075     0.536 0.000
nodecov.Wealth    0.011 1.011     0.005 0.024
```

manually estimating edges odds ratios

```
exp(-2.594929)
```

```
[1] 0.07465118
```

Odds Interpretation

edges 0.075

An increase of 1 in the number of edges increases the log odds with -2.595 and it makes the odds of the effect taking place 0.075 times larger.

OR < 1 means that the odds of forming edges in this network is small.

nodecov.Wealth 0.011

An increase of 1 in wealth increases the log odds with 0.011 and it makes the odds of the effect taking place 1.011 times larger.

OR > 1 means that the odds of wealth influencing marriage are bigger

OR = 1 means there is little association between wealth influencing marriage.

Probabilities

$exp(\theta)/(1 + exp(\theta))$

```
snafun::stat_ef_int(m, type = "prob")
```

```
               Estimate  Prob Std.Error  Pval
edges            -2.595 0.069     0.536 0.000
nodecov.Wealth    0.011 0.503     0.005 0.024
```

manually estimating edges probabilities

```
exp(-2.594929)/ (1 + exp(-2.594929))
```

```
[1] 0.0694655
```

Probabilities Interpretation

```
0.069 * 100
```

```
[1] 6.9
```

```
0.503 * 100
```

```
[1] 50.3
```

6.9 % prob of forming edges

small prob, so it is consistent with the OR result

50.3 % prob that wealth predicts marriage

it's a bit more than 1/2 so it is consistent with the OR result

7 steps Research Flow

Let's organize our research in steps

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1. Literature research/ Questions & Hypotheses (based on the literature) [- data collection]
2. Data Exploration + terms selection
3. Running models (baseline, exploratory models, final model)
4. MCMC diagnostics of the best selected model
5. Goodness of fit
6. Interpreting results
7. Write up about your research telling how this journey went

TIP: Rely on these steps for your report

See you on Next Week!

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